

SIMATS ENGINEERING

ASSESSMENT TOOL 3

NAME: N.Vijetha

REG.NO: 192311258

COURSE NAME: CSA1603

COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 20%

QUESTIONS: 05

Total Marks:50

Annexure A

COMPARITIVE ANALYSIS

<i>Criteria</i>	Scope of Items (2)	Comparison Depth(2.5)	Use of Evidence (2)	Critical Thinking (2)	Clarity of Presentation (1)	<i>Total(10)</i>
<i>Weightage</i>	20%	25%	20%	20%	10%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I _N.Vijetha / 192311258_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

N.Vijetha

RUBRICS FOR CALCULATION:

		Comparative analysis			
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

Q1 . Dataset: Assessment Tool 3: Comparative Analysis (25% → 5 Marks)

Time duration :45 Minutes
Assessment Tool 3 (Low)

Q1 . Dataset: Customer Data

Customer Income (₹) Spending Score

C1	30000	60
C2	60000	40
C3	50000	70
C4	28000	65
C5	65000	35

Question:

Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability. (BL3 – 1 Mark)

Q1

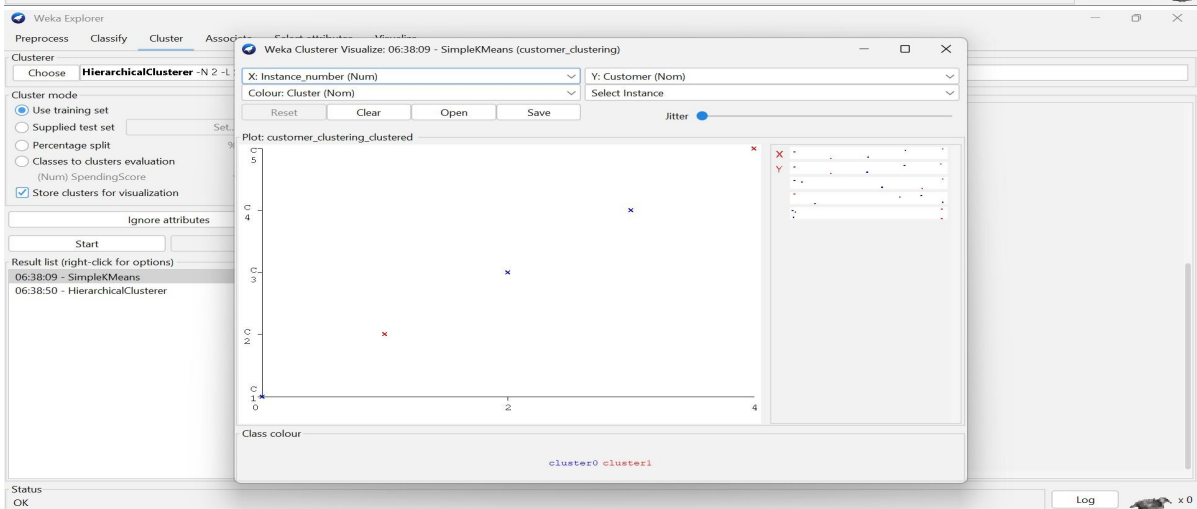
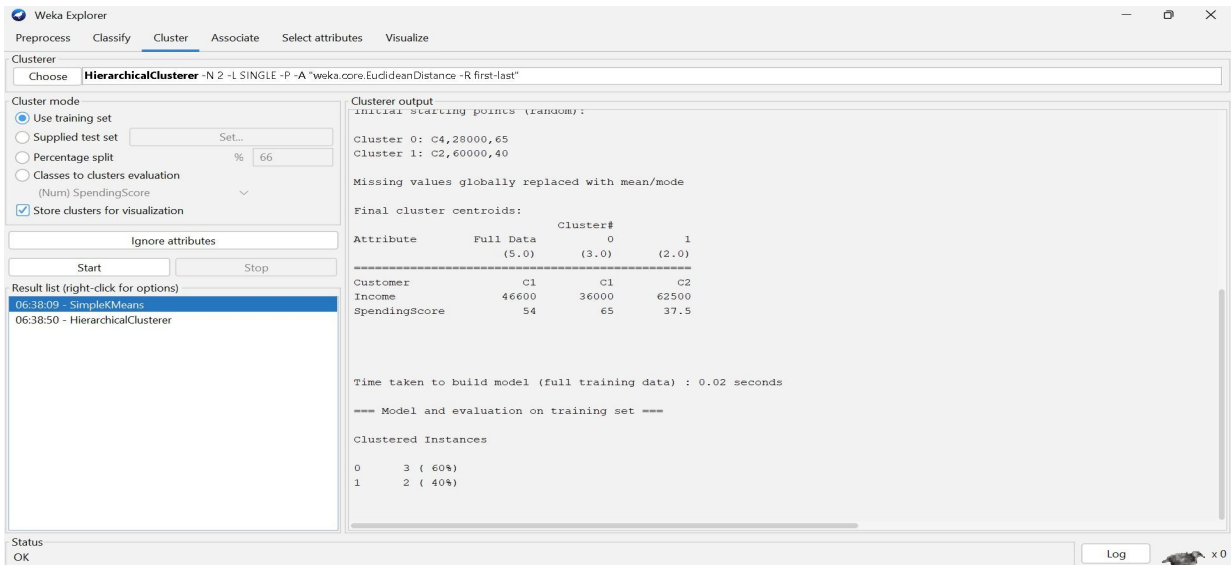
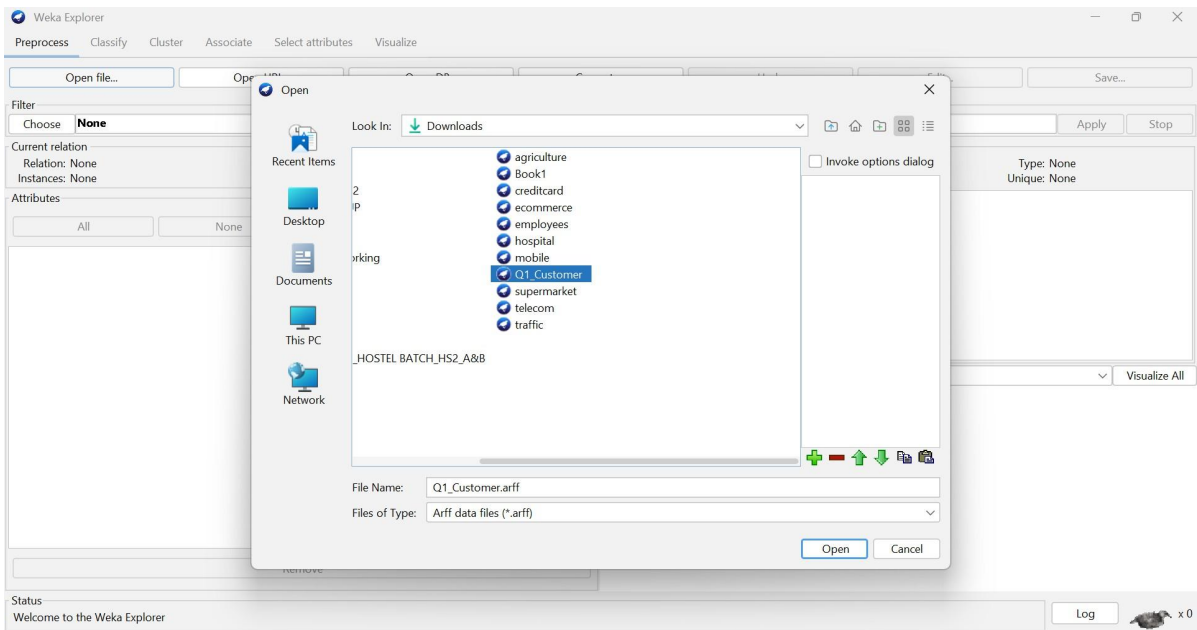
K-MEANS VS HIERARCHICAL CLUSTERING

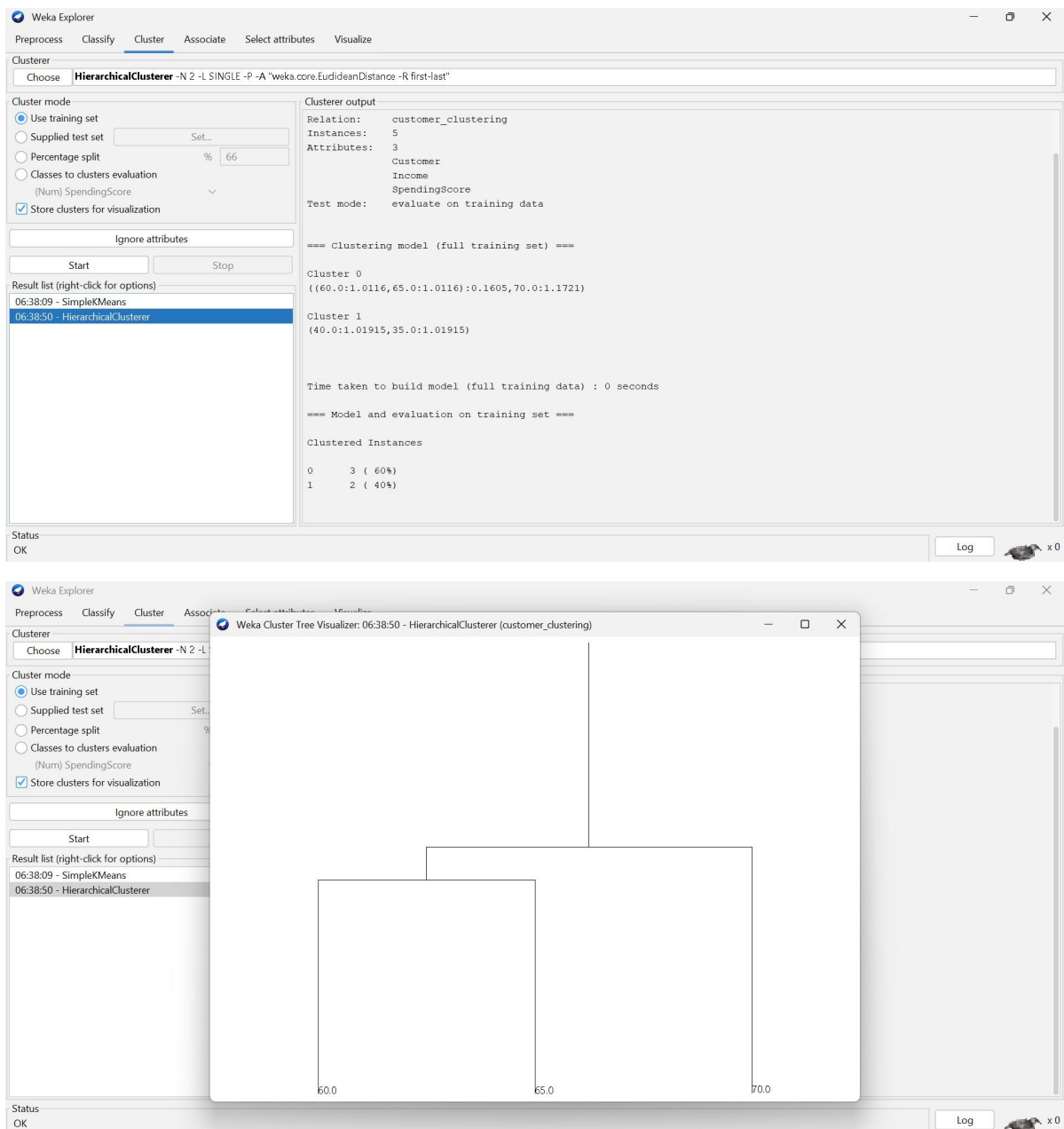
K-means:
This clustering divides the customers into a fixed no. of clusters using centroids.
⇒ It is fast and works well for large datasets
⇒ It repeatedly assigns customers to the nearest centroid and updates the centroid.

Hierarchical clustering:
This clustering creates clusters step-by-step
⇒ It is slower & requires more computation for large datasets
⇒ It uses a dendrogram to show relationships between customers.

Conclusion:
→ k-means is better for large datasets customer data
→ while hierarchical clustering is useful for understanding cluster relationships.

OUTPUT:





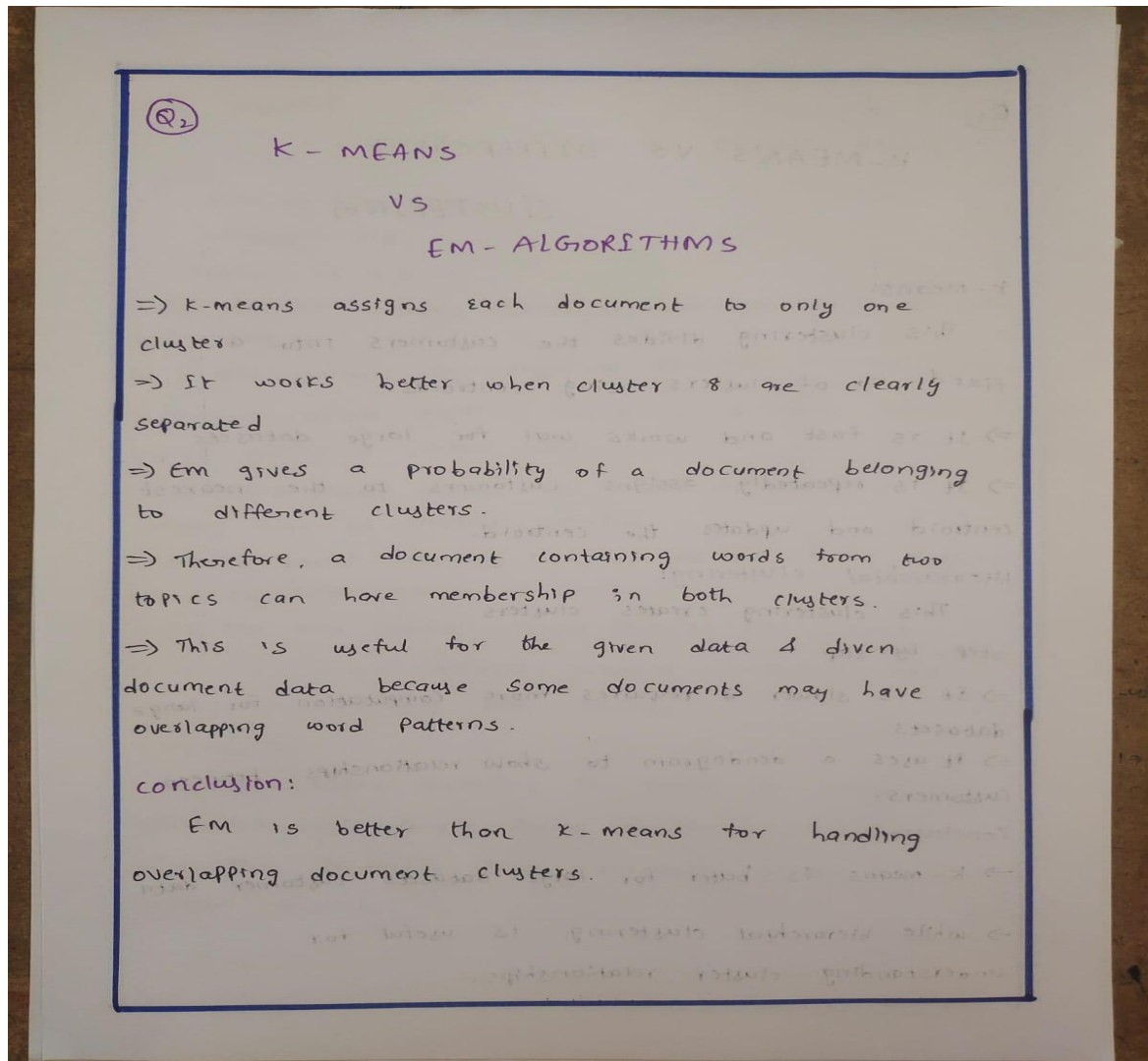
Q2. Dataset: Document Data

Document Word1 Word2 Word3

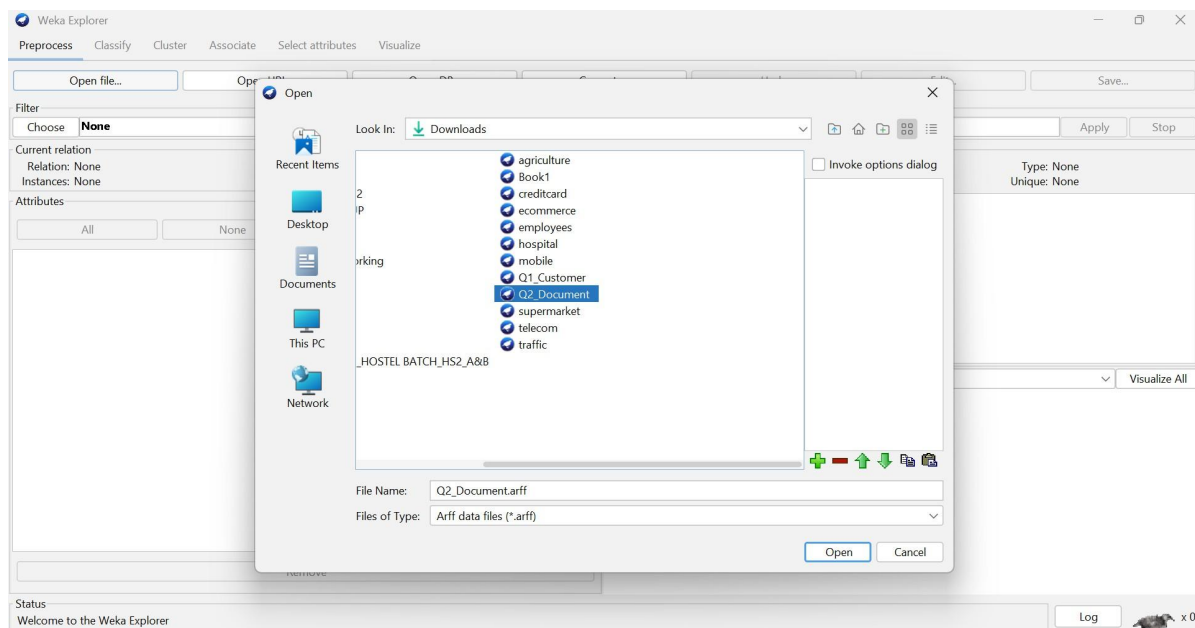
D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

Question:

Compare K-means and EM algorithms in handling overlapping clusters for the given dataset.
(BL4 – 1 Mark)



OUTPUT:



Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer: Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -i 500 -num-slots 1 -S 10

Cluster mode

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split
- ☐ Classes to clusters evaluation
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

- 06:43:03 - SimpleKMeans

Clusterer output

```
Cluster 0: D4,0.75,0.15,0.1
Cluster 1: D2,0.1,0.6,0.3

Missing values globally replaced with mean/mode

Final cluster centroids:
Attribute    Full Data    Cluster#
              (5.0)      (2.0)      (3.0)
=====
Document     D1           D3           D1
Word1         0.41         0.775        0.1667
Word2         0.35         0.125        0.5
Word3         0.24         0.1          0.3333

Time taken to build model (full training data) : 0.01 seconds

=== Model and evaluation on training set ===

Clustered Instances

0    2 ( 40%)
1    3 ( 60%)
```

Status: OK

Log

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer: Choose **EM** -I 100 -N -1 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 100

Cluster mode

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split
- ☐ Classes to clusters evaluation
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

- 06:43:03 - SimpleKMeans
- 06:43:27 - EM

Clusterer output

```
D4      4
D5      2
[total] 10
Word1
mean     0.41
std. dev. 0.3007

Word2
mean     0.35
std. dev. 0.1949

Word3
mean     0.24
std. dev. 0.12

Time taken to build model (full training data) : 0.01 seconds

=== Model and evaluation on training set ===

Clustered Instances

0    5 (100%)

Log likelihood: -0.90915
```

Status: OK

Log

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer: Choose **EM** -I 100 -N -1 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 100

Cluster mode

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split
- ☐ Classes to clusters evaluation
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

- 06:43:03 - SimpleKMeans
- 06:43:27 - EM

Clusterer output

```
=== Run information ===

Scheme:      weka.clusterers.EM -I 100 -N -1 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 1
Relation:    document_clustering
Instances:   5
Attributes:  4
              Document
              Word1
              Word2
              Word3

Test mode:   evaluate on training data

=== Clustering model (full training set) ===

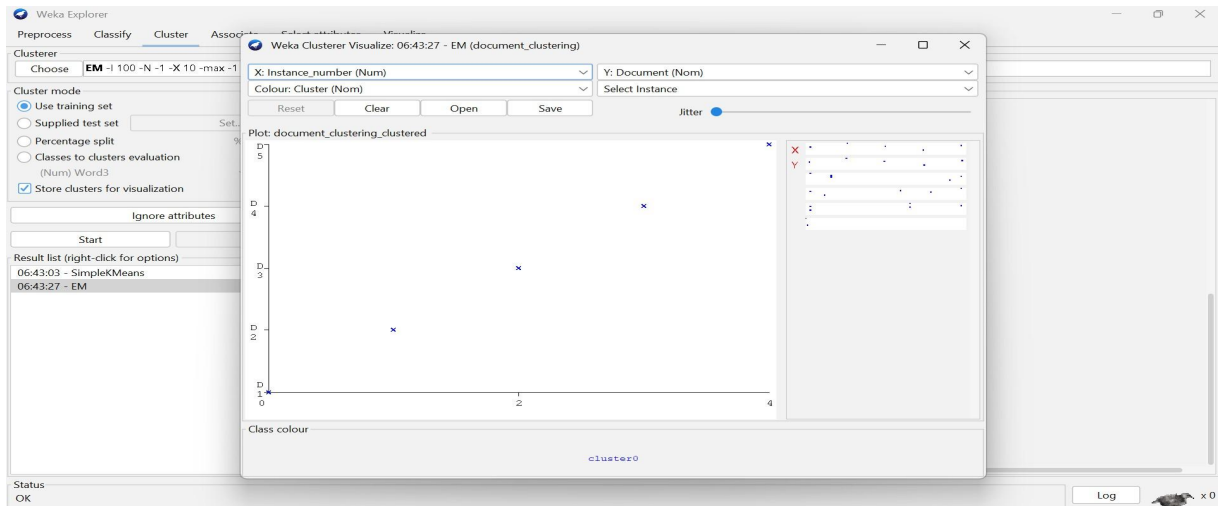
EM
==

Number of clusters selected by cross validation: 1
Number of iterations performed: 2

Attribute    Cluster
              (1)
=====
Document     ~
              ~
```

Status: OK

Log



Q3. Dataset: Geographic Data

Point Latitude Longitude

P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

Question:

Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. (BL4 – 1 Mark)

Q3

GEOGRAPHIC DATA

Euclidean vs Manhattan

Given,

P₁ - P₅ have latitude values from 12.97 - 13.10

longitude values from 80.20 - 80.30

- Euclidean distance measures the direct straight line distance between two geographic points.

Formula:

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

- Manhattan distance adds the absolute latitude & longitude differences.

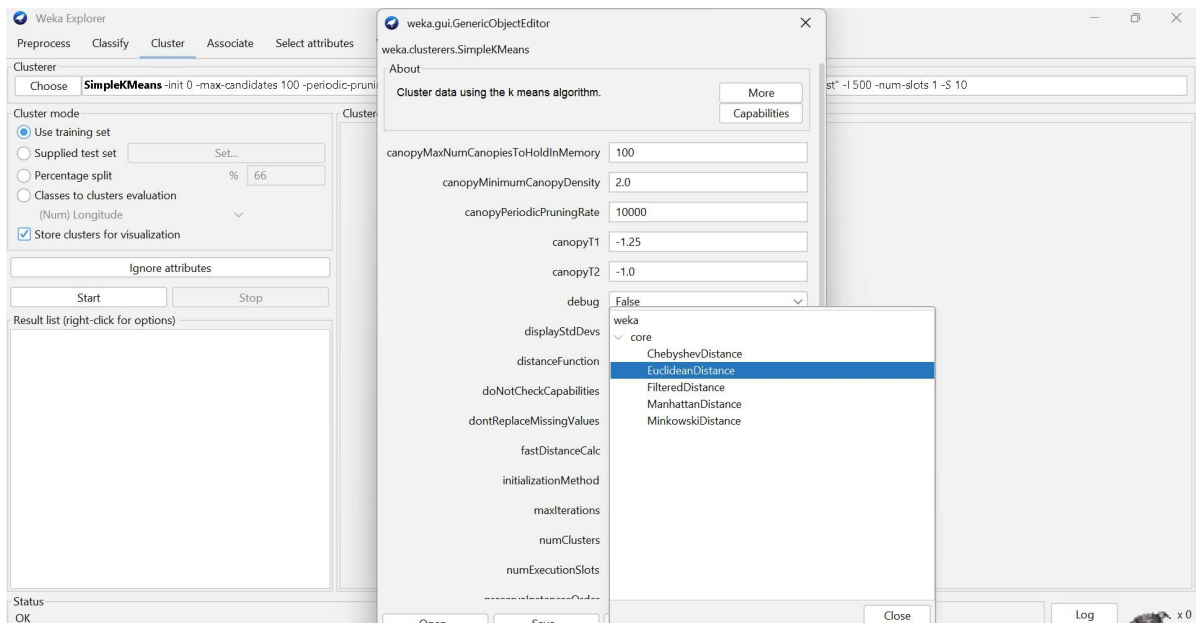
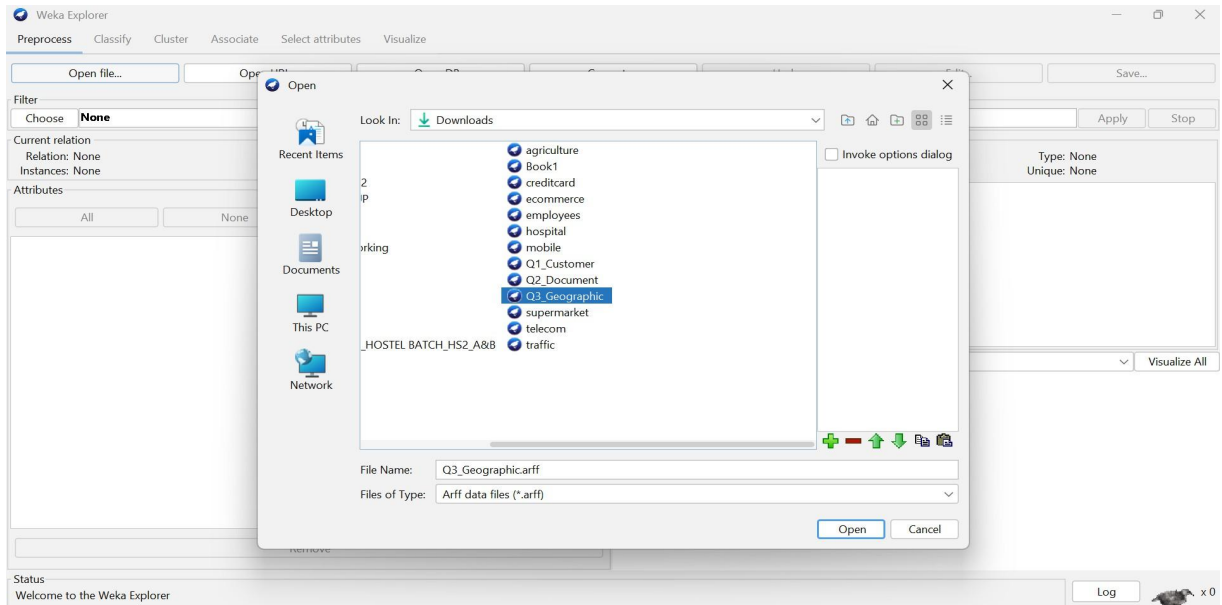
Formula:

$$d = |x_1 - x_2| + |y_1 - y_2|$$

Conclusion:

Euclidean distance is more suitable here because the data represents geographic locations and direct spatial closeness is most important.

OUTPUT:



Weka Explorer

Preprocess **Cluster** Associate Select attributes Visualize

Clusterer: Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode:
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) Longitude
☒ Store clusters for visualization

Ignore attributes: []
 Start Stop

Result list (right-click for options):
 06:48:47 - SimpleKMeans

Cluster output:

```
Initial starting points (random):
Cluster 0: P4,13.05,80.3
Cluster 1: P2,13.1,80.25

Missing values globally replaced with mean/mode

Final cluster centroids:
Attribute      Full Data      Cluster#
              (5.0)         0         1
              (5.0)         (2.0)        (3.0)
-----
Point          P1          P3          P1
Latitude       13.04      12.985     13.0767
Longitude      80.248     80.21      80.2733

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0      2 ( 40%)
1      3 ( 60%)
```

Status: OK Log x 0

Weka Explorer

Preprocess **Cluster** Associate Select attributes Visualize

Clusterer: Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode:
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) Longitude
☒ Store clusters for visualization

Ignore attributes: []
 Start Stop

Result list (right-click for options):
 06:48:47 - SimpleKMeans

weka.gui.GenericObjectEditor

weka.clusterers.SimpleKMeans

About

Cluster data using the k means algorithm.

canopyMaxNumCanopiesToHoldInMemory: 100
 canopyMinimumCanopyDensity: 2.0
 canopyPeriodicPruningRate: 10000
 canopyT1: -1.25
 canopyT2: -1.0
 debug: False
 displayStdDevs: weka
 distanceFunction: core
 doNotCheckCapabilities: ChebyshevDistance
 dontReplaceMissingValues: EuclideanDistance
 fastDistanceCalc: FilteredDistance
 initializationMethod: ManhattanDistance
 maxIterations: MinkowskiDistance
 numClusters: []
 numExecutionSlots: []
 preserveInstancesOrder: []

Close Log x 0

Weka Explorer

Preprocess **Cluster** Associate Select attributes Visualize

Clusterer: Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.ManhattanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode:
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) Longitude
☒ Store clusters for visualization

Ignore attributes: []
 Start Stop

Result list (right-click for options):
 06:48:47 - SimpleKMeans
 06:49:48 - SimpleKMeans

Cluster output:

```
Initial starting points (random):
Cluster 0: P4,13.05,80.3
Cluster 1: P2,13.1,80.25

Missing values globally replaced with mean/mode

Final cluster centroids:
Attribute      Full Data      Cluster#
              (5.0)         0         1
              (5.0)         (1.0)        (4.0)
-----
Point          P1          P4          P1
Latitude       13.05      13.05      13.04
Longitude      80.25      80.3       80.235

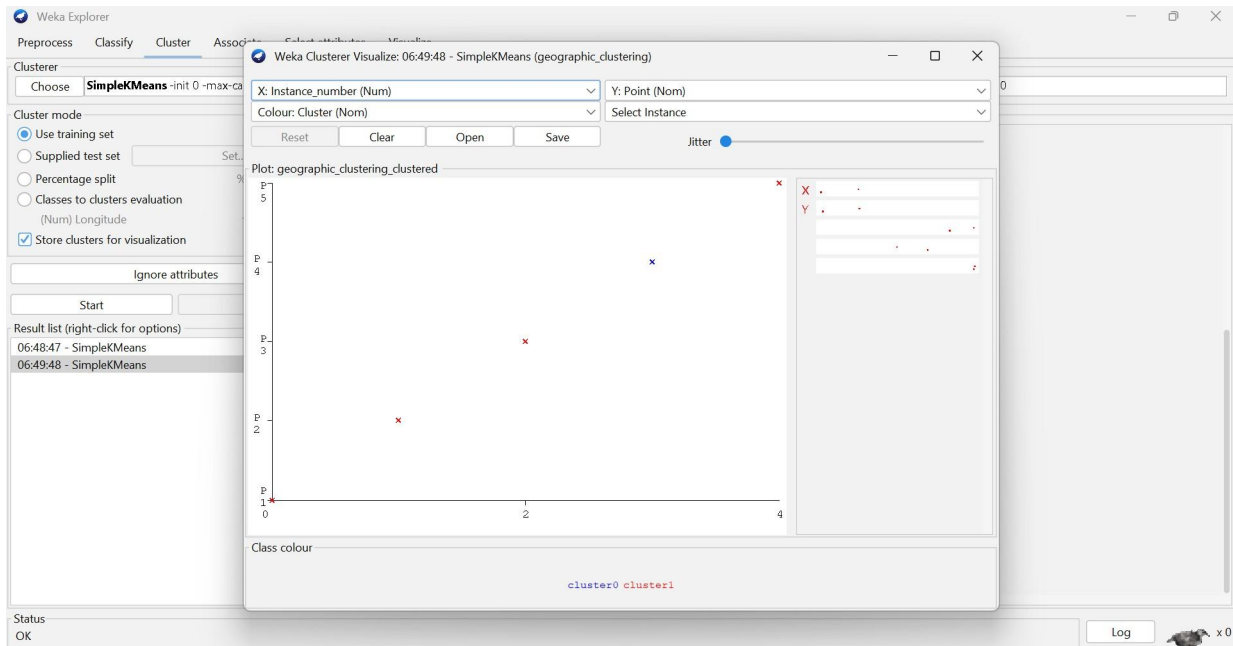
Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0      1 ( 20%)
1      4 ( 80%)
```

Status: OK Log x 0



Q4. Dataset: Retail Products

Product Sales (₹) Frequency

P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

Question:

Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. (BL3 – 1 Mark)

Q4

RETAIL PRODUCTS

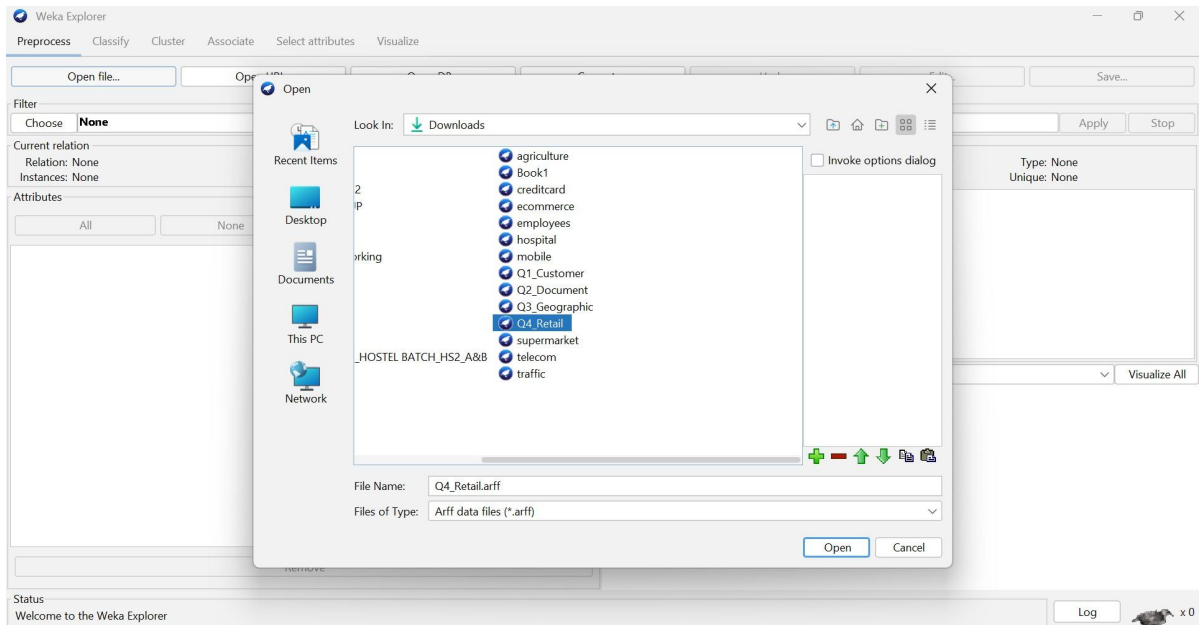
↳ AGGLOMERATIVE
VS
DIVISIVE

GIVEN,
Sales = ₹20,000 - ₹52,000 and frequency
⇒ 7-22

- Agglomerative clustering starts with each product as a separate cluster
- It gradually merges similar products based on sales and frequency
- For example, P₂ (₹50,000, 22) & P₄ (₹52,000, 12) have high sales and can be considered relatively similar.
- Divisive clustering starts with all five products in one cluster.
- It then splits the products into similar groups.
- Agglomerative clustering is simple and suitable for grouping.

⇒ These retail products based on sales and the purchase frequency.

OUTPUT:



Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer
Choose **HierarchicalClusterer** -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"

Cluster mode
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) Frequency
☒ Store clusters for visualization

Ignore attributes
Start Stop

Result list (right-click for options)
 06:53:39 - HierarchicalClusterer

Cluster output

```

Relation: retail_products
Instances: 5
Attributes: 3
Product
Sales
Frequency
Test mode: evaluate on training data

=== Clustering model (full training set) ===

Cluster 0
(20.0:1.01286,22.0:1.01286)

Cluster 1
((10.0:1.01286,12.0:1.01286):0.01412,8.0:1.02699)

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0    2 ( 40%)
1    3 ( 60%)
  
```

Status
OK

Log

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer
Choose **MakeDensityBasedClusterer** -M 1.0E-6 -W weka.clusterers.SimpleKMeans --init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -r

Cluster mode
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) Frequency
☒ Store clusters for visualization

Ignore attributes
Start Stop

Result list (right-click for options)
 06:53:39 - HierarchicalClusterer
 06:54:50 - MakeDensityBasedClusterer

Cluster output

```

Discrete Estimator. Counts = 1 2 1 2 2 (Total = 8)
Attribute: Sales
Normal Distribution. Mean = 29000 StdDev = 2943.9203
Attribute: Frequency
Normal Distribution. Mean = 10 StdDev = 1.633

Cluster: 1 Prior probability: 0.4286

Attribute: Product
Discrete Estimator. Counts = 2 1 2 1 1 (Total = 7)
Attribute: Sales
Normal Distribution. Mean = 51000 StdDev = 1000
Attribute: Frequency
Normal Distribution. Mean = 21 StdDev = 1

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0    3 ( 60%)
1    2 ( 40%)

Log likelihood: -12.6953
  
```

Status
OK

Log

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer
Choose **MakeDensityBasedClusterer**

Cluster mode
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) Frequency
☒ Store clusters for visualization

Ignore attributes
Start Stop

Result list (right-click for options)
 06:53:39 - HierarchicalClusterer
 06:54:50 - MakeDensityBasedClusterer

Weka Clusterer Visualize: 06:54:50 - MakeDensityBasedClusterer (retail_products)

X: Instance_number (Num) Y: Product (Nom)
 Colour: Cluster (Nom) Select Instance
 Reset Clear Open Save Jitter

Plot: retail_products_clustered

Class colour
cluster0 cluster1

Status
OK

Log

Q5 . Dataset: Mixed Attributes

Item Price (₹) Category Rating

I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

Question:

Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. (BL4 – 1 Mark)

Q5

MIXED ATTRIBUTES

- CENTROID BASED VS DENSITY-BASED

Given,

Price $\Rightarrow$ ₹250 - ₹550

category = A/B,

Rating = 3.5 - 4.7

$\Rightarrow$ The dataset contains numerical attributes such as price and rating.

$\Rightarrow$ It also contains a categorical attributes, category (A/B).

$\Rightarrow$ centroid-based clustering, such as k-means forms groups around a central point.

$\Rightarrow$ k-means mainly works with numerical values, so category must be properly encoded.

$\Rightarrow$ density-based clustering such as DBSCAN forms clusters based on dense groups of products.

$\Rightarrow$ It can also identify unusual products as outliers.

$\Rightarrow$ for this mixed dataset, suitable encoding and distance measurement are required.

Conclusion:

centroid-based clustering is simple for numerical attributes, while density-based clustering is useful when the products form irregular groups (or) contain outliers.

OUTPUT:

